

Module: Corporate Finance & Equity

Valuation

School of Economics

University of Cape Town

Valuation of listed companies Using the

Free Cash Flow to the Firm Method

Learning Objectives

- ❑ Understand the Discounted Cash Flow Valuation (DCF)
- ❑ Understand how to estimate the inputs into DCF valuation
- ❑ Understand some of the adjustments necessary to perform a DCF Valuation
- ❑ Understand the meaning of “free cash flow to the firm” and “free cash flow to equity” and how they differ.
- ❑ Employ the Free Cash Flow Model to value the firm (enterprise value) and ordinary shares of the firm approach.

Discounted Cashflow Valuation (DCF)

- ❑ What is it?: In discounted cash flow valuation, the value of an asset is the present value of the expected cash flows on the asset.
- ❑ Philosophical Basis: Every asset has an intrinsic value that can be estimated, based on its characteristics in terms of cash flows, growth and risk.
- ❑ Information Needed: To use discounted cash flow valuation, you need:
 - to estimate the life of the asset
 - to estimate the cash flows during the life of the asset
 - to estimate the discount rate to apply to these cash flows to get present value
- ❑ Market Inefficiency: Markets are assumed to make mistakes in pricing assets across time, and are assumed to correct themselves over time, as new information comes out about assets.

Advantages of DCF Valuation

- ❑ Since DCF valuation, if done right, is based upon an asset's fundamentals, it should be less exposed to market moods and perceptions.
- ❑ If good investors buy businesses, rather than stocks (the Warren Buffet adage), discounted cash flow valuation is the right way to think about what you are getting when you buy an asset.
- ❑ DCF valuation forces you to think about the underlying characteristics of the firm, and understand its business. If nothing else, it brings you face to face with the assumptions you are making when you pay a given price for an asset.
- ❑ If you buy into the notion of value being driven by a company's cash flows, you are immunized (to the extent that you have a long time horizon) from what the market thinks about your investment.

Disadvantages of DCF valuation

- ❑ Since it is an attempt to estimate intrinsic value, it requires far more explicit inputs and information than other valuation approaches
- ❑ These inputs and information are not only noisy (and difficult to estimate), but can be manipulated by the analyst to provide the conclusion he or she wants. The quality of the analyst then becomes a function of how well he or she can hide the manipulation.
- ❑ In an intrinsic valuation model, there is no guarantee that anything will emerge as under or over valued. Thus, it is possible in a DCF valuation model, to find every stock in a market to be over valued. This can be a problem for:
 - equity research analysts, whose job it is to follow sectors and make recommendations on the most under and over valued stocks in that sector
 - equity portfolio managers, who have to be fully (or close to fully) invested in equities

When DCF Valuation works best

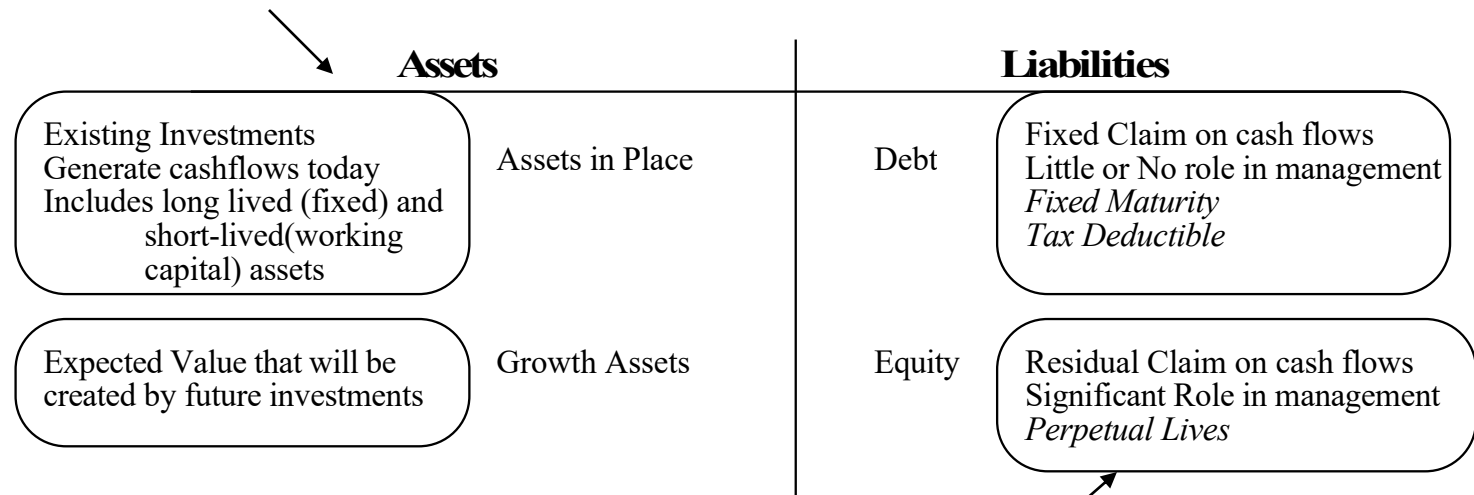
- ❑ At the risk of stating the obvious, this approach is designed for use for assets (firms) that derive their value from their capacity to generate cash flows in the future.
- ❑ It follows that investments that do not generate cash flows (gold, currencies, collectibles) cannot be valued.
- ❑ It works best for investors who:
 - have a long time horizon, allowing the market time to correct its valuation mistakes and for price to revert to “true” value or
 - are capable of providing the catalyst needed to move price to value, as would be the case if you were an activist investor or a potential acquirer of the whole firm
 - are not easily swayed or affected by peer pressure, i.e., market movements that are contrary to their “value” views

The essence of intrinsic value

- ❑ In intrinsic valuation, you value an asset based on its fundamentals (or intrinsic characteristics).
- ❑ For cash flow generating assets, the intrinsic value will be a function of the magnitude of the expected cash flows on the asset over its lifetime and the uncertainty about receiving those cash flows.
- ❑ Discounted cash flow valuation is a tool for estimating intrinsic value, where the expected value of an asset is written as the present value of the expected cash flows on the asset, with either the cash flows or the discount rate adjusted to reflect the risk.

DCF Choices: Equity Valuation versus Firm Valuation

Firm Valuation: Value the entire business



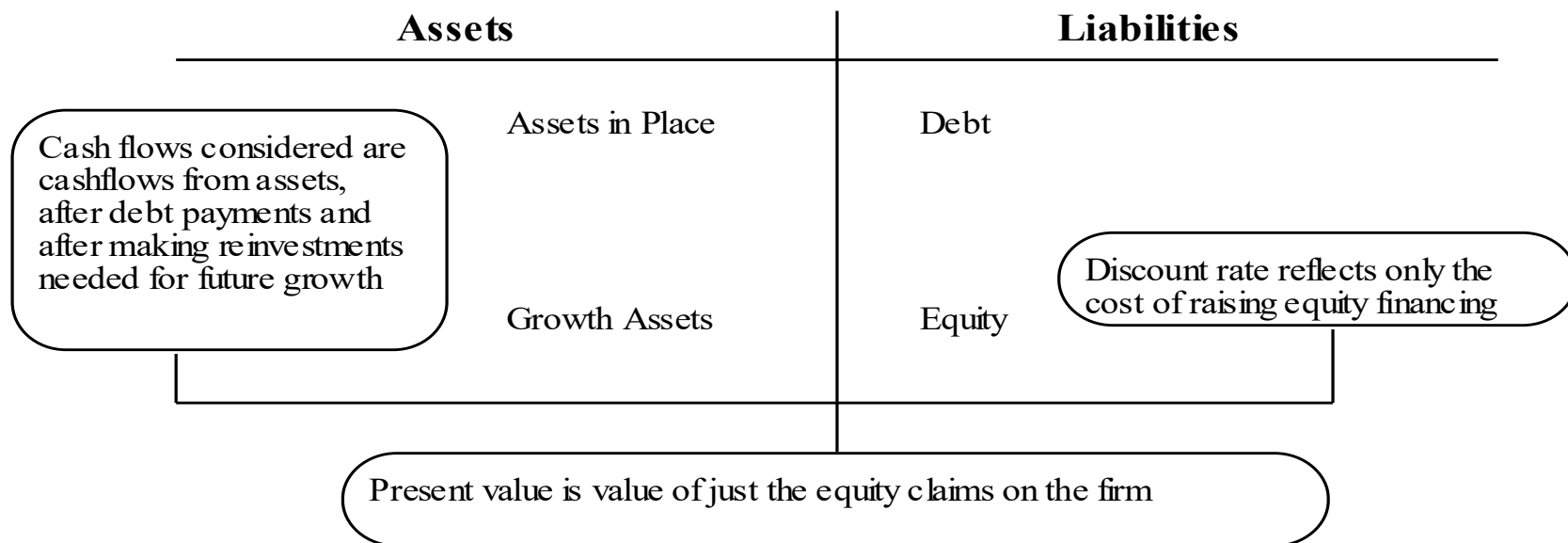
Equity valuation: Value just the equity claim in the business

The value of a business and the value of equity can not only be very different numbers but they can move in different directions...

- A firm like GE has a value for its business that is vastly in excess of its value for equity, because of the substantial debt that it carries. In contrast, the value of the business and the value of equity for a firm like Apple that has no debt will converge.
- You can have valuable businesses, where the equity is worth nothing because the firm has borrowed too much....

Equity Valuation

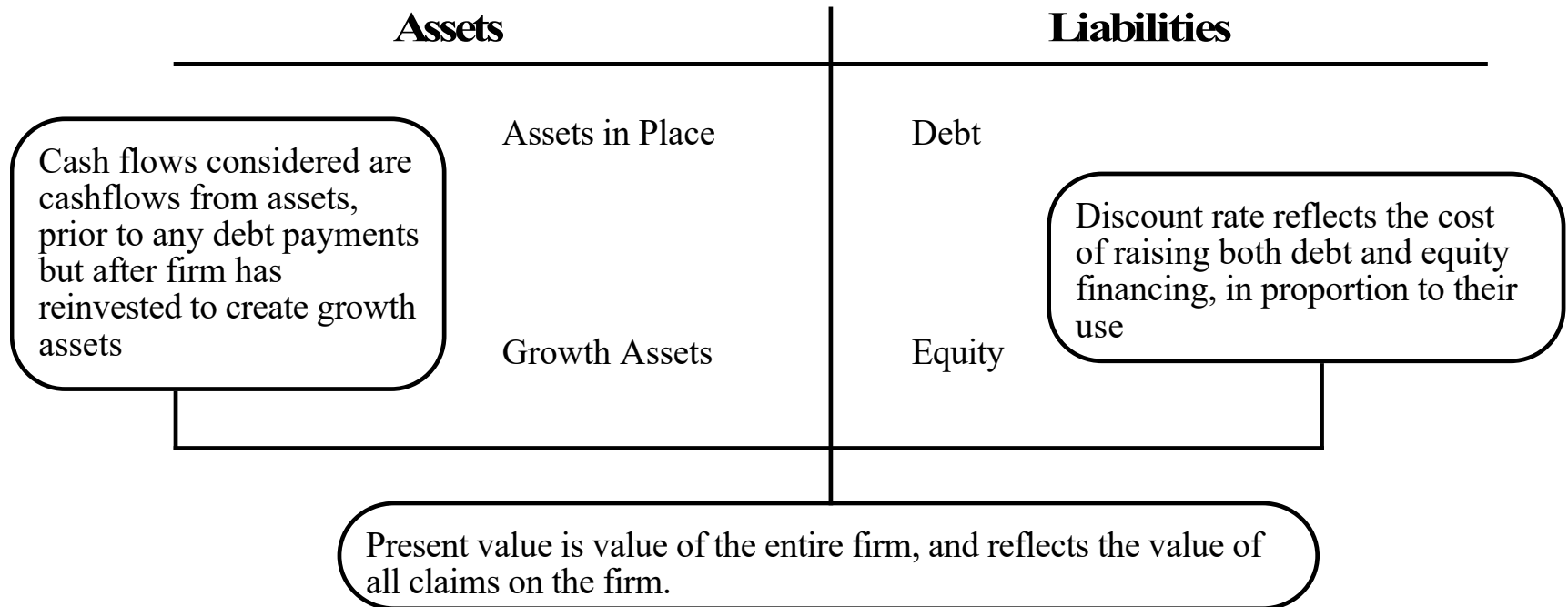
Figure 5.5: Equity Valuation



- ❑ The value of equity is the present value of cash flows to the equity investors discounted back at the rate of return that those equity investors need to make to break even (the cost of equity).
- ❑ In the strictest sense of the word, the only cash flow stockholders in a publicly traded firm get from their investment is dividends, and the dividend discount model is the simplest and most direct version of an equity valuation model.

Firm Valuation

Figure 5.6: Firm Valuation



- ❑ A firm includes not just the equity, but all claim holders. The cash flow to the firm is the collective cash flow that all claim holders make from the firm, and it is discounted at the weighted average of their different costs.
- ❑ Valuing a business is counter intuitive, because it does require the owner of a business to think of the bank that lent money to the business as a partner. The cash flow to the business is the sum of the cash flows to the owner (equity) and the cash flows to debt (interest and principal payments).

Cash Flows and Discount Rates

Example 1

Assume that you are analysing a company with the following cashflows for the next five years.

Year	CF to Equity	Interest Expense (1-t)	CF to Firm
1	R 50	R 57,6	R 107,6
2	R 60	R 57,6	R 117,6
3	R 68	R 57,6	R 125,6
4	R 76.2	R 57,6	R 133.8
5	R 83.49	R 57,6	R 141.09
Terminal Value R 1603.0			R 5289.5893

- ☐ Assume also that the cost of equity is 13.625% and the firm can borrow long term at 10%. (The tax rate for the firm is 28%). Assume that the sustainable growth rate after the fifth year is 8%.
- ☐ The current market value of equity is R1,073 and the value of debt outstanding is R800.
 - Assume that the above cashflows as given. Further assume that the debt outstanding is a perpetual bond with fixed interest payments in each period.
 - In addition, assume that the costs of equity and debt are also given as shown above.

Treat these cashflows as given. In the pages to come, we will talk about estimating these cashflows in detail. Note that we are assuming a perpetual bond with fixed interest payments in each period. The costs of equity and debt are also given.

Equity versus Firm Valuation

- ❑ Method 1: Discount CF to Equity at Cost of Equity to get value of equity
 - Cost of Equity = 13.625%
 - Value of Equity = $50/1.13625 + 60/1.13625^2 + 68/1.13625^3 + 76.2/1.13625^4 + (83.49+1603)/1.13625^5 = \mathbf{R1073,01}$

- ❑ Method 2: Discount CF to Firm at Cost of Capital to get value of firm
 - Cost of Debt = Pre-tax rate (1- tax rate) = 10% (1-0,28) = 7,2%
 - Cost of Capital = 13.625% (1073/1873) + 7,2% (800/1873) = 10,8807%

 - PV of Firm = $107.6/1.108807 + 117.6/1.108807^2 + 125.6/1.108807^3 + 133.8/1.108807^4 + (141.09+5289.5893)/1.108807^5 = \mathbf{R3613,57}$

 - Value of Equity = Value of Firm - Market Value of Debt
 $= \mathbf{R\ 3564,18 - R\ 800 = R2813,57}$

The two values can only converge if the market values of equity and debt for the firm happen to be equal to the estimated values of equity and debt. If this does not hold, as in this case, the weights you use for your cost of capital calculation - which are based on current market value - will not be consistent with the debt ratio you are assuming for estimating for your Free Cash Flows to Equity calculations and the two values will diverge.

The reason the two values converge is because the market values of equity and debt for this firm happen to be equal to the estimated values of equity and debt. If this does not hold, the weights you use for your cost of capital calculation - which are based upon current market value - will not be consistent with the debt ratio you are assuming for estimating for your FCFE calculations and the two values will diverge.

First Principle of Valuation

❑ Discounting Consistency Principle: Never mix and match cash flows and discount rates.

❑ Mismatching cash flows to discount rates is deadly. Examples of such problems- in red.

- Discounting cashflows after debt cash flows (equity cash flows) at the weighted average cost of capital will lead to an upwardly biased estimate of the value of equity
- Discounting pre-debt cashflows (cash flows to the firm) at the cost of equity will yield a downward biased estimate of the value of the firm.

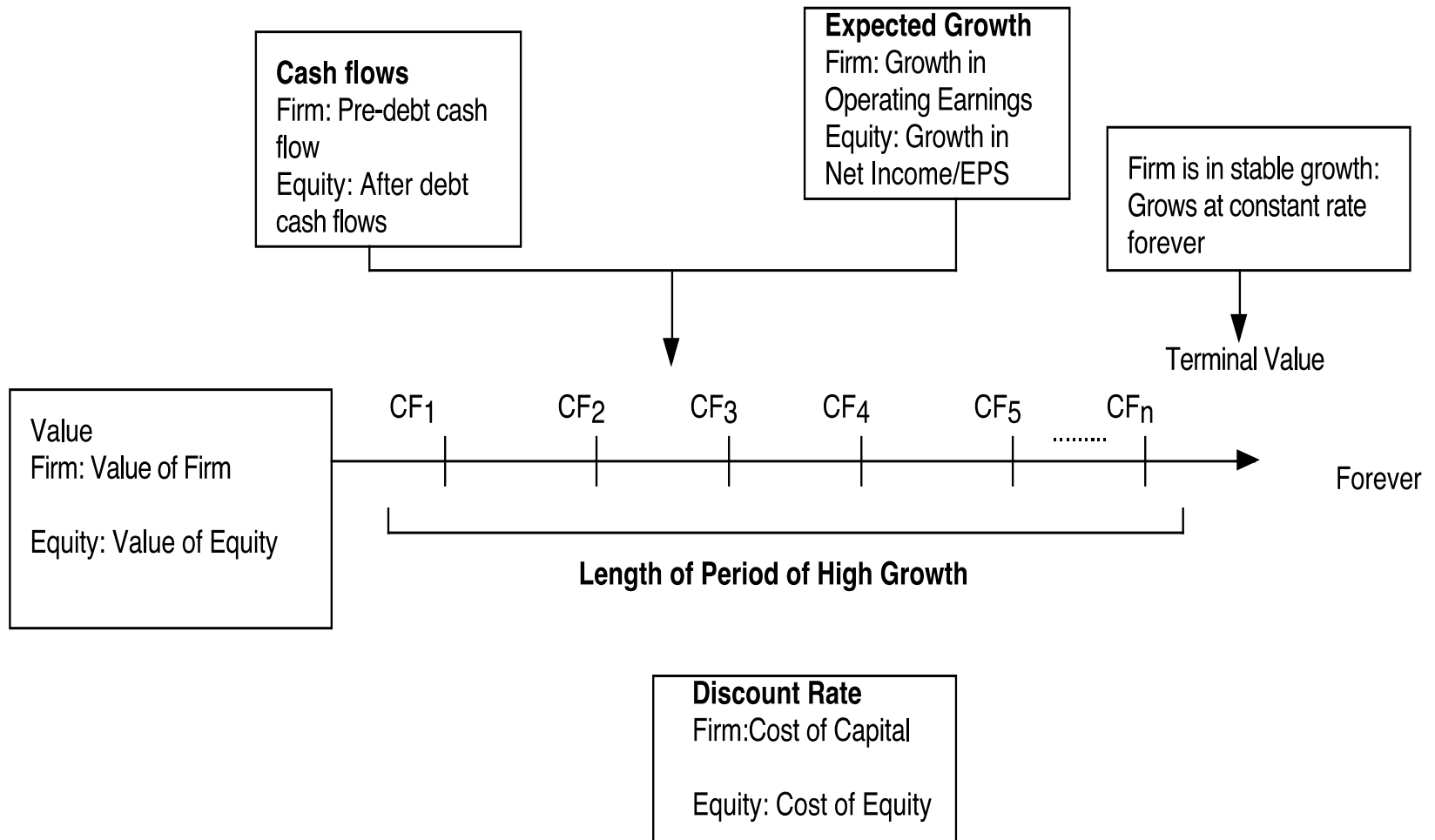
NB. It is a principle that is often violated because we often talk about discount rates and cashflows loosely and costs of capital and equity interchangeably. We need to be more precise about words like free cash flow (to whom? The firm? Equity?) and discount rates to avoid these problems.

Discounted Cash Flow Valuation: The Steps

1. Estimate the discount rate or rates to use in the valuation
 - Discount rate can be either a cost of equity (if doing equity valuation) or a cost of capital (if valuing the firm)
 - Discount rate can be in nominal terms or real terms, depending upon whether the cash flows are nominal or real
 - Discount rate can vary across time.
2. Estimate the current earnings and cash flows on the asset, to either equity investors (CF to Equity) or to all claimholders (CF to Firm)
3. Estimate the future earnings and cash flows on the firm being valued, generally by estimating an expected growth rate in earnings.
4. Estimate when the firm will reach “stable growth” and what characteristics (risk & cash flow) it will have when it does.
5. Choose the right DCF model for this asset and value it.

Generic DCF Valuation Model

DISCOUNTED CASHFLOW VALUATION



Generic DCF Valuation Model

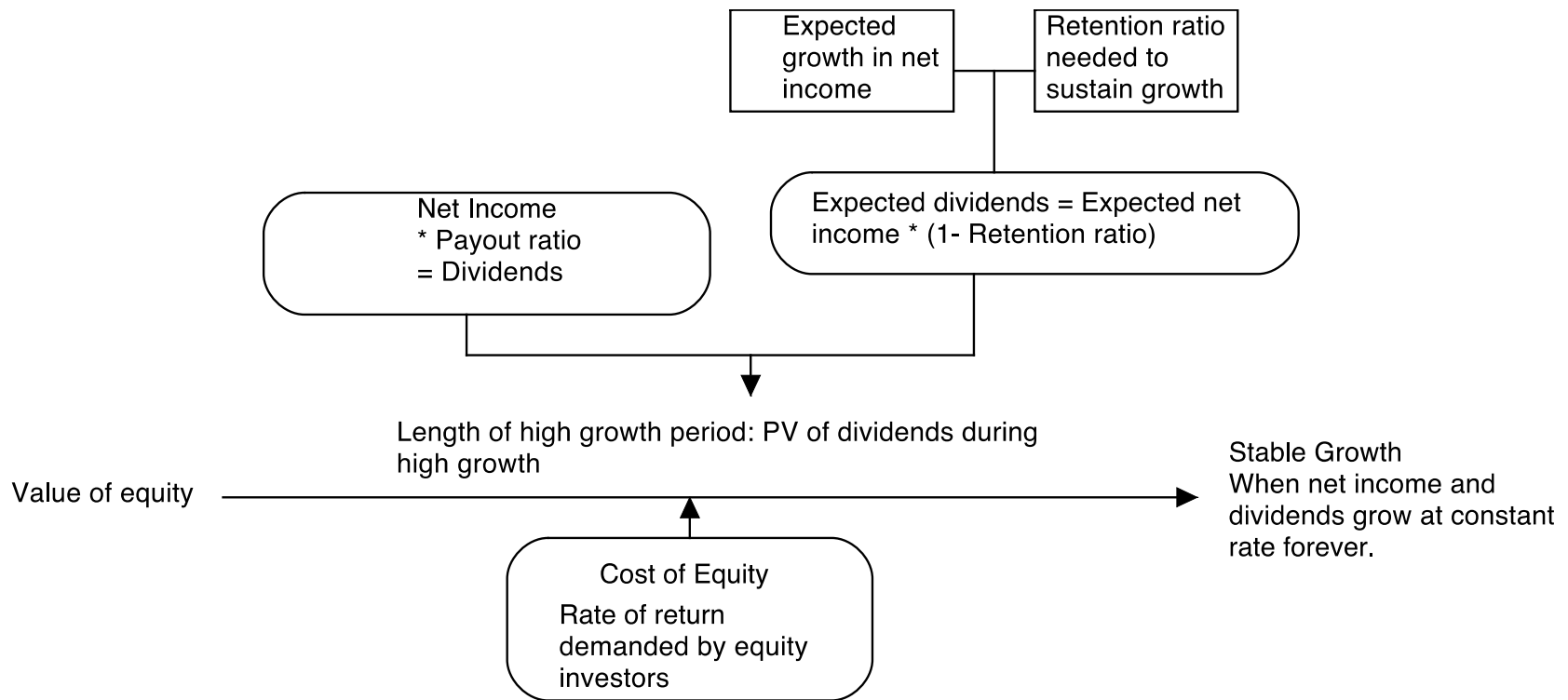
❑ The four pillars of value:

- Cashflows
- Potential for high growth
- Length of the high growth period (before the firm starts growing at the same rate as the economy)
- Discount rate

❑ Note the differences and the need for consistency:

- With equity -> Cashflows to equity - > **Growth rate in net income -> Discount at the cost of equity**
- With firm -> Cashflows to firm - > **Growth rate in operating income -> Discount at the cost of capital**

Start easy: The Dividend Discount Model



The dividend discount model is the oldest DCF model around and it reflects the fact that bonds preceded equities into financial markets. Thus, investors in publicly traded stocks initially considered them to be “special” bonds, with the dividends replacing coupons and an uncertain (and appreciating) principal replacing the face value of the bond. The biggest weakness of the dividend discount model is not that it cannot value stocks that don’t pay dividends but that it misvalues companies that pay out more or less than they can afford to in dividends.

Valuing the entire business: The FCFF model

Free Cash Flow to the Firm

Operating Income after Tax given by $(EBIT * (1-t) = NOPAT$
 add. Depreciation & Ammortisation
 Less: Capital Expenditure (CAPEX)
 Less: Changes in None Cash Working Capital
 = Free Cash Flows to the Firm (FCFF)

Expected growth in
Operating Income

Re-investment needed to
sustain growth

Expected Free Cash Flows to the Firm =
 Expected operating income $*$ (1- Re-investment rate)

Value of Operating Assets
 + cash & None operating Assets
 - Debt
 = Value of Equity

Length of High Growth Period: PV of FCFF during high growth period

Stable growth
 When operating income &
 FCFF grow at a constant rate
 forever

Cost of Capital
 Weighted Average of costs of Equity
 and Debt (WACC)

Valuing the entire business: The FCFF model

- ❑ FCFF models differ from FCFE models as they focus on valuing the entire business, rather than just the equity stake in the business.
- ❑ Rather than focus on just cash flows to equity, FCFF measures cash flow to all capital providers. It is a pre-debt cash flow.
- ❑ The discount rate has to reflect the weighted average of what all capital providers demand for investing and is thus the cost of capital.
- ❑ Since the cash flows are operating cash flows, you have to add on the value of cash and non-operating assets to get to the value of the firm.
- ❑ Subtracting debt from the value of the firm yields the value of equity in the business.

Estimating Inputs: Discount Rates

- ❑ At an intuitive level, the discount rate used should be consistent with both the riskiness and the type of cashflow being discounted.
 - Equity versus Firm: If the cash flows being discounted are cash flows to equity, the appropriate discount rate is a cost of equity. If the cash flows are cash flows to the firm, the appropriate discount rate is the cost of capital.
 - Currency: The currency in which the cash flows are estimated should also be the currency in which the discount rate is estimated.
 - Nominal versus Real: If the cash flows being discounted are nominal cash flows (i.e., reflect expected inflation), the discount rate should be nominal.

Recently, I came across a valuation of a Mexican company, where the cashflows were in nominal pesos but the discount rate used was the dollar cost of capital of a U.S. acquirer.... As a result, the value was inflated by more than 300%....

Issues to note from the previous slide

- ❑ While discount rates are a critical ingredient in discounted cashflow valuation, we tend to spend far too much time on discount rates and far too little on cashflows. The most significant errors in valuation are often the result of failures to estimate cash flows correctly....
- ❑ As companies increasingly become global, and multiple listings abound, the consistency principle becomes very important. The currency used in estimating cash flows should also be the currency in which you estimate discount rates - dollar discount rates for dollar cashflows and rand discount rates for rand cash flows.

Steps in Cash Flow Estimation

- ❑ Estimate the current earnings of the firm
 - If looking at cash flows to equity, look at earnings after interest expenses - i.e. net income
 - If looking at cash flows to the firm, look at operating earnings after taxes
- ❑ Consider how much the firm invested to create future growth
 - If the investment is not expensed, it will be categorized as capital expenditures. To the extent that depreciation provides a cash flow, it will cover some of these expenditures.
 - Increasing working capital needs are also investments for future growth
- ❑ If looking at cash flows to equity, consider the cash flows from net debt issues (debt issued - debt repaid)

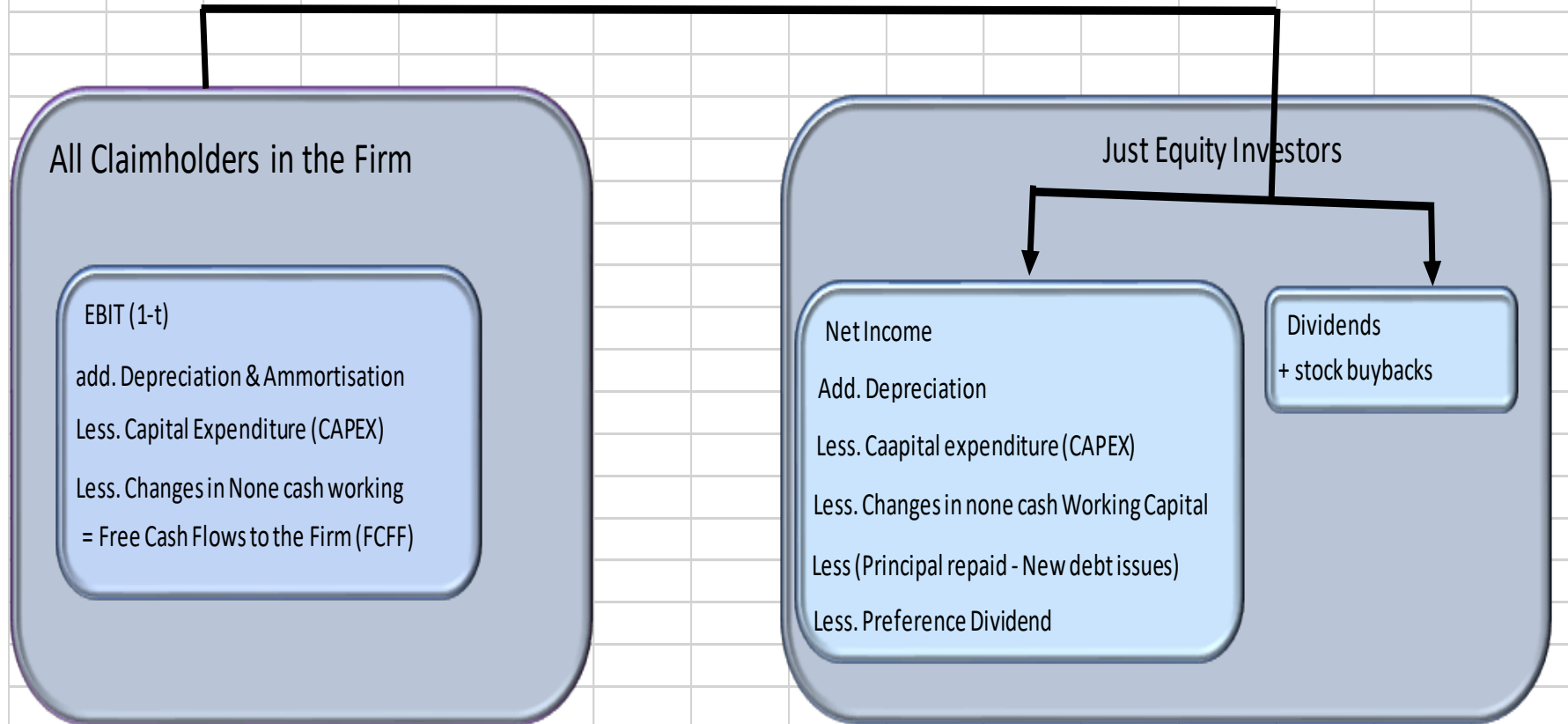
Sets up the process. Much as we do not trust accounting statements, that is where we go for the information. Accounting earnings are not cash flows.

We do not draw a distinction between discretionary and required capital expenditures. Once you introduce growth in earnings, the distinction becomes largely symbolic.

Draws a distinction between cashflows to equity and the firm again.

Measuring Cash Flows

Cash flows can be measured to



Lays out the three definitions of cashflows. The strictest measure is the dividend measure. (In fact, there are some who do not count stock buybacks.) The more expansive equity measure is the free cashflow to equity, which you can think off as potential dividends. The free cashflow to the firm is the cash available for all claimholders in the firm - it is before cashflows to any of the claimholders in the firm - debt or equity.

Dealing with Operating Lease Expenses

- ❑ Operating Lease Expenses are treated as operating expenses in computing operating income. In reality, operating lease expenses should be treated as financing expenses, with the following adjustments to earnings and capital:
 - ❑ Debt Value of Operating Leases = Present value of Operating Lease Commitments at the pre-tax cost of debt
 - ❑ When you convert operating leases into debt, you also create an asset to counter it of exactly the same value.
 - ❑ Adjusted Operating Earnings
 - $\text{Adjusted Operating Earnings} = \text{Operating Earnings} + \text{Operating Lease Expenses} - \text{Depreciation on Leased Asset}$
- As an approximation, this works:
- $\text{Adjusted Operating Earnings} = \text{Operating Earnings} + \text{Pre-tax cost of Debt} * \text{PV of Operating Leases}.$

Dealing with Operating Lease Expenses

- ❑ Computational note: You use the pre-tax cost of debt as your discount rate because you are discounting pre-tax operating lease expenses.
- ❑ From an intuitive standpoint, there is little difference between a term loan (where you pay off a loan in equal annual installments) and an operating lease. It may be more like unsecured debt than secured debt but it is debt.
- ❑ It is not just debt that is affected when you convert operating leases to debt. The operating income also will change.

Operating Leases at Padua in 2019

- Assume that Padua has conventional debt of about R 1.97 billion on its balance sheet and its pre-tax cost of debt is about 6%. Its stated operating income is R1, 012 million. Its operating lease payments in 2019 were R978 million and its commitments for the future are below:

Year	Commitment (millions)	Present Value (at 6%)
1	R899.00	R848.11
2	R846.00	R752.94
3	R738.00	R619.64
4	R598.00	R473.67
5	R477.00	R356.44
6&7	R982.50 each year	R1,346.04

- Debt Value of leases = R4,396.85 (Also value of leased asset)
- Debt outstanding at Padua = R1,970 m + R4,397 m = R6,367 m
- Adjusted Operating Income = Stated Operating Income + Operating Lease expense this year - Depreciation
 $= \text{R1,012 m} + 978 \text{ m} - 4397 \text{ m} / 7 = \text{R1,362 million (7 year life for assets)}$
- Approximate Operating Income = R1,012 m + R 4397 m (.06) = R1,276 m

The Gap reports its lease commitments in its financial statements. The present value of operating lease expenses is computed using the pre-tax cost of debt. (An argument can be made that the unsecured cost of debt should be used.)

The Collateral Effects of Treating Operating Leases as Debt

Assuming Book value of equity is R3130 and tax rate of 35%

Conventional Accounting	Operating Leases Treated as Debt								
<i>Income Statement</i> EBIT& Leases = 1,990 - Op Leases = 978 EBIT = 1,012	<i>Income Statement</i> EBIT& Leases = 1,990 - Deprecn: OL= 628 EBIT = 1,362 Interest expense will rise to reflect the conversion of operating leases as debt. Net income should not change.								
<i>Balance Sheet</i> Off balance sheet (Not shown as debt or as an asset). Only the conventional debt of R1,970 million shows up on balance sheet	<i>Balance Sheet</i> <table><tr><td>Asset</td><td></td><td>Liability</td><td></td></tr><tr><td>OL Asset</td><td>4397</td><td>OL Debt</td><td>4397</td></tr></table> Total debt = 4397 + 1970 = R6,367 million	Asset		Liability		OL Asset	4397	OL Debt	4397
Asset		Liability							
OL Asset	4397	OL Debt	4397						
Cost of capital = 8.20%(7350/9320) + 4% (1970/9320) = 7.31% Cost of equity for Padua ltd = 8.20% After-tax cost of debt = 4% Market value of equity = 7350	Cost of capital = 8.20%(7350/13717) + 4% (6367/13717) = 6.25%								
Return on capital = 1012 (1-0.35)/(3130+1970) = 12.90%	Return on capital = 1362 (1-0.35)/(3130+6367) = 9.30%								

Traces the effect of converting operating leases to debt. Both operating income and capital invested increase. The net effect on return on capital will depend upon which increases more.

In general, treating leases as debt will affect the return spread you earn (by changing both the cost of capital and return on capital). If the spread decreases (as it does for the Gap), your values will decrease when you capitalize leases. If the spread increases, your value will increase. In either case, the values that you get with the capitalized leases is the more realistic estimate of value.

R&D Expenses: Operating or Capital Expenses

- ❑ Accounting standards require us to consider R&D as an operating expense even though it is designed to generate future growth. It is more logical to treat it as capital expenditures.
- ❑ To capitalize R&D,
 - Specify an amortizable life for R&D (2 - 10 years)
 - Collect past R&D expenses for as long as the amortizable life
 - Sum up the unamortized R&D over the period. (Thus, if the amortizable life is 5 years, the research asset can be obtained by adding up $\frac{1}{5}$ th of the R&D expense from five years ago, $\frac{2}{5}$ th of the R&D expense from four years ago...:etc,

The argument used by accountants - that R&D yields uncertain benefits - is specious. You could make the same argument about other investments - investing in a factory in an emerging market, deciding to build a concept car - and you are forced to treat these as capital expenditures.

To capitalize R&D, you need to specify

- On average, how long it takes between the time you do research and a commercial product emerges from the research. (This is the amortizable life)
- R&D expenses from the past (for a period equivalent to the amortizable life). If your firm has not been in existence for that long, you would go back for as many years as you can.
- Depreciation schedules - stick with the simplest which is straight line depreciation.

Capitalizing R&D Expenses: SAP

R & D was assumed to have a 5-year life. (amounts are in Euros)

Year	R&D Expense	Unamortized	Amortization this year	
Current	1020.02	1.00	1020.02	
-1	993.99	0.80	795.19	R 198.80
-2	909.39	0.60	545.63	R 181.88
-3	898.25	0.40	359.30	R 179.65
-4	969.38	0.20	193.88	R 193.88
-5	744.67	0.00	0.00	R 148.93
Value of research asset		=	R 2,914 million	
Amortization of research asset in 2019		=	R 903 million	
Increase in Operating Income = 1020 - 903 =			R117 million	

- The amortizable life is an assumption based upon SAP's business. It would be shorter in other businesses (such as computer chips) and longer in businesses that need regulatory approval (such as pharmaceuticals).
- Note that the amortization is 1/5 of the R&D expense each year.
- We are also assuming that R&D expenses are spent at the end of each year - not realistic, but simplifies analysis - that is why there is no amortization of the current year's expense.
- The effect of capitalizing R&D will be greatest at firms where R&D is growing over time and be non-existent at firms with flat R&D.

Value of Research asset = 1020.02+ 795,19+545,63+359,30+193,88+0 = 2,914

Amortization of research = 198,80+181,88+179,65+193,88+148,93 = 903

The Effect of Capitalizing R&D at SAP

Assuming book value of debt is R530million and a tax rate of 36.54%

<i>Conventional Accounting</i>	<i>R&D treated as capital expenditure</i>				
<i>Income Statement</i> EBIT& R&D = 3045 - R&D = 1020 EBIT = 2025 EBIT (1-t) = 1285 m	<i>Income Statement</i> EBIT& R&D = 3045 - Amort: R&D = 903 EBIT = 2142 (Increase of 117 m) EBIT (1-t) = 1359 m Ignored tax benefit = $(1020-903)(0.3654) = 43$ Adjusted EBIT (1-t) = $1359+43 = 1402$ m (Increase of 117 million) Net Income will also increase by 117 million				
<i>Balance Sheet</i> Off balance sheet asset. Book value of equity at R3,768 million is understated because biggest asset is off the books.	<i>Balance Sheet</i> <table> <tr> <td>Asset</td> <td>Liability</td> </tr> <tr> <td>R&D Asset 2914</td> <td>Book Equity +2914</td> </tr> </table> Total Book Equity = $3768+2914= 6782$ mil	Asset	Liability	R&D Asset 2914	Book Equity +2914
Asset	Liability				
R&D Asset 2914	Book Equity +2914				
<i>Capital Expenditures</i> Conventional net cap ex of R2 million	<i>Capital Expenditures</i> Net Cap ex = $2+ 1020 - 903 = 119$ mil				
<i>Cash Flows</i> EBIT (1-t) = 1285 - Net Cap Ex = 2 FCFF = 1283	<i>Cash Flows</i> EBIT (1-t) = 1402 - Net Cap Ex = 119 FCFF = 1283 m				
Return on capital = $1285/(3768+530)$ = 29.90%	Return on capital = $1402/(6782+530)$ = 19.93%				

Traces out the effects of capitalizing R&D. The key aspect is that operating income will be increased by exactly the same amount that net capital expenditures will be increased, with the increase being:

Change in operating income and net cap ex = Current year's R&D expense - R&D amortization

In other words, the free cashflow to the firm will not change as a result of this capitalization. So why do it? It allows us to

- Get a better sense of the profitability of the firm (operating income and return on capital)
- Better estimate how much the firm is reinvesting for future growth.
- Get a measure of whether R&D is value creating or value destroying.

Now we Turn our attention to the FCFF Method

This is our main Focus

The steps

What are the steps?

Do we still remember?

FCFF Valuation: The Calculations

FIRM OR ORDINARY SHARE VALUATION USING THE FCF to the FIRM MODEL REQUIRES THE FOLLOWING STEPS:

- **STEP 1: If not given, calculate WACC.** This requires the calculation of:
 - Cost of equity (k_e).
 - Cost of preference shares (k_p).
 - After tax cost of debt (k_d)
 - Appropriate weights of the CS (based on target CS or FVs of invested capital). **(Refer to: Cost of Capital Module).**
- **STEP 2: If not given, calculate (on the valuation date) the PV, FairV or MV of debt, pref. shares and non-operating assets.** (Refer to: Basic Valuation Module).
- **STEP 3: If not given, prepare forecast income statement and balance sheet for each year of an explicit forecast period using past trends and relationships (5 to 10 years in practice).** This is the period of unstable or 'super' growth.
- **STEP 4: If not given, calculate the investment in NWC and CAPEX (both excluding non-operating assets, e.g. Investments and excess cash).**

FCFTF Valuation: Calculations – Cont'd

- **STEP 5: If not given, calculate the expected future FCFTF for each year of the explicit forecast period (determined in Step 3).**
The free cash flow to the firm (FCFTF) is calculated as follows:

	Explicit Forecast Period		
	Year 1	Year 2	Year 3
	R	R	R
EBIT and B4 non-operating income	XX	XX	XX
Less: Taxation (28% x EBITABNOI)	(XX)	(XX)	(XX)
NOPAT [EBITABNOI x (1 – t)]	XX	XX	XX
Add: Depreciation & amortization	XX	XX	XX
Gross free cash flow to the firm	XX	XX	XX
Less: Investment in NWC	(XX)	(XX)	(XX)
Less: Investment in CAPEX	(XX)	(XX)	(XX)
FREE CASH FLOW TO THE FIRM	XX	XX	XX

- **STEP 6: Calculate the continuing value using the constant growth model.** This is the FCFTF in the final year of the explicit period multiply by $(1 + g)$ and then divide by $(WACC - g)$. E.G. $CV = FCFTF_3(1 + g)/(WACC - g)$.

FCFTF Valuation: Calculations – Cont'd

- **STEP 7:** Calculate the PV of the FCFTF for each year in the explicit forecast period and the Continuing Value (CV) using WACC as the discount rate.
- **STEP 8:** Calculate the value of the firm (enterprise value), being the sum of: (1) the PV of the FCFF for the explicit forecast period; (2) the PV of the CV; and (3) the Fair Value/Market Value of the non-operating assets (E.g.).
- **STEP 9:** Calculate the value to ordinary shareholders, being enterprise value less Market Value of debt and Market Value of preference shares.
 - To get value per ordinary share, divide the value to ordinary shareholders by the number of ordinary shares in issue.
 - **How do you determine whether the share is over- or under-valued?**

Free Cash Flow Valuation Model

Example 2

The following information has been determined for Pleasure Ltd:

- The WACC for the company is 20%, the fair value of the debt today ($t = 0$) is equal to R1.45m and the cash balance in excess of operational requirements today ($t = 0$) equals R0.75m. The free cash flow to the firm has been calculated for the next three years as follows:

	Explicit Forecast Period		
	Year 1	Year 2	Year 3
	Rm	Rm	Rm
EBIT and B4 non-operating income	2.50	3.00	3.50
Less: Taxation (28% x EBITABNOI)	(0.70)	(0.84)	(0.98)
NOPAT [EBITABNOI x (1 – t)]	1.80	2.16	2.52
Add: Depreciation & amortization	0.40	0.60	0.80
Gross free cash flow to the firm	2.20	2.76	3.32
Less: Investment in NWC	(0.30)	(0.40)	(0.50)
Less: Investment in CAPEX	(0.50)	(0.60)	(0.70)
FREE CASH FLOW TO THE FIRM	1.40	1.76	2.12

- From year 4 and onwards, the FCFTF are expected to grow at a constant rate of 5% per year. **REQUIRED:** Calculate the value of each ordinary share using the free cash flow to the firm model. Number of ordinary shares in issue is 400 000.

Free Cash Flow Valuation Model

Example 2 – Suggested Solution

- FIRST**, calculate the PV of FCFTF for the explicit forecast period and the PV of the CV.

	Explicit Forecast Period		
	Year 1	Year 2	Year 3
	Rm	Rm	Rm
Free cash flow to the firm	1.40	1.76	2.12
Continuing value (see Wkg)			14.84
FCFTF with CV	1.40	1.76	16.96
PVIF (at 20%)	0.8333	0.6944	0.5787
PV of Future FCFTF	1.16662	1.22214	9.81475

WORKING:

$$\text{Continuing value} = P_3 = \text{FCFTF}_3(1+g)/(\text{WACC} - g)$$

$$\text{Continuing value} = P_3 = 2.12(1.05)/(0.20 - 0.05) = \text{R}14.84\text{m}$$

Free Cash Flow Valuation Model

Example 2 – Suggested Solution – Cont'd

- **SECOND, calculate:** (a) the value of the firm (enterprise value); (b) the value of ordinary shares; and (c) the value per ordinary share as follows:

	Rm
PV of free cash flows to the firm (1.17 + 1.22 + 9.81)	12.20m
Add: Fair value of non-operating assets: Excess cash	<u>0.75</u>
Enterprise value	12.95m
Less: Market value of debt	<u>(1.45)</u>
Value of ordinary shares	11.50m
Value per ordinary share (R11.50m/400 000)	<u>R28.75</u>

- **SEE Example 3**

Free Cash Flow Valuation Model

Lecture Example 3 – Suggested Solution

- Calculating the net investment in CAPEX (for each year):
- Use the Net book values (i.e. carrying amount) column of PPE (excluding None Operating Assets):
 - Opening Balance + Purchases – Depreciation = Closing Balance
 - **Purchases = Closing Balance + Depreciation – Opening Balance**

	20X3	20X4	20X5
	Rm	Rm	Rm
Closing balance (CA of NCAs)	80	100	110
Add: Depreciation	8	12	14
	88	112	124
Less: Opening balance	(50)	(80)	(100)
Net investment in CAPEX	38	32	24

Free Cash Flow Valuation Model

Example 3 – Suggested Solution – Cont'd

- Calculating investment in net working capital (for each year):
- If not given, first calculate the net working capital per year (excluding None Operating Assets) and then the cash outflow (INCREASE) or cash inflow (DECREASE) for each year of the explicit forecast period:

	20X3	20X4	20X5	20X6
	Rm	Rm	Rm	Rm
Inventory	35	40	50	65
Accounts receivable	25	30	45	60
Accounts payable	(4)	(6)	(14)	(13)
Tax payable	(1)	(3)	(4)	(5)
Net working capital	55	61	77	107
Change in NWC – cash out		6	16	30

Free Cash Flow Valuation Model

Example 3 – Suggested Solution – Cont'd

- Calculating the FCFTF and the PV of FCFTF for the explicit forecast period and for the continuing value:

	20X4	20X5	20X6
	Rm	Rm	Rm
EBIT x (1 – t)	86.4	118.8	140.4
Add: Depreciation	8	12	14
Gross operating cash flow	94.4	130.8	154.4
Less: Net investment in CAPEX	(38)	(32)	(24)
Less: Net investment in WC	(6)	(16)	(30)
Free cash flow to the FIRM	50.4	82.8	100.4
CV [100.4(1.06)]/(0.18 – 0.06)			886.9
FCFTF with CV	50.4	82.8	987.3
PVIF at 18%	0.8475	0.7182	0.6086
PV of FCFTF with CV	42.7	59.5	600.9

Free Cash Flow Valuation Model

Example 3 – Suggested Solution – Cont'd

- Calculating the value of a 75% shareholding in TNT (Pty) Ltd:

	Rm
PV of FCFTF (42.7 + 59.5 + 600.9)	703.1
Add: Non-operating assets:	
Fair Value of other investments (2m x R12.00)	24.0
Cash and cash equivalents	5.0
Enterprise value	732.1
Less: Fair value of borrowings (58 +1)	(59.0)
Value of ordinary shares	673.1
FV of a 75% interest (75% x R673.1m)	504.8

Free Cash Flow Valuation Model

Example 3 –POINTS TO NOTE

Common Assumption: Depreciation = W&T Allowance

- This assumption is meant to simplify the calculation of NOPAT using the EBIT approach as shown below:
 - $\text{NOPAT} = \text{EBIT} \times (1 - 0.28).$

Free Cash Flow Valuation Model

Lecture Example 3 –POINTS TO NOTE

Valuation model based on FREE CASH FLOW TO EQUITY

- FCFTE is less popular than FCFTF – Hence, the reason why it is not discussed in detail in this module.
- The two models (FCFTF and FCFTE) differ in two main respects, namely, how free cash flows are calculated and the discount rate used.
 - FCFTE is determined after deducting interest expense and preference share dividends.
 - FCFTE (including CV) is discounted using the COST OF EQUITY and **NOT** WACC.